

# Q4 – pLDDT Values and Fragmentation of AlphaFold Reconstructions at the Scale of the *H. sapiens* Proteome: Reproduction and Arity Cross-Correlation Extension

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**Abstract**—We reproduce and extend the pLDDT-fragmentation analysis (Q0/Q4) of Cazals & Sarti [1] at the scale of the human (*H. sapiens*) proteome using AlphaFold Database v4 predictions. The analysis is grounded in a path-graph filtration of residues ordered by decreasing predicted local-distance difference test (pLDDT) confidence score, from which a per-protein persistence diagram is derived using Union-Find with the Elder Rule. Five statistics are computed from each diagram: the fraction of positive-persistence pairs  $f_{cp}^+$ , mean persistence  $\bar{p}$ , normalised persistence entropy  $H_p$ , the peak connected-component count  $N_{cc}^{max}$ , and the count of persistent local maxima  $PLM(t_p)$ .

From 23,391 total fragments we analyse 16,787 first-fragment (F1) entries with  $n \geq 200$  residues, recovering a Pearson correlation of  $r = 0.928$  between  $f_{cp}^+$  and  $H_p$  (paper target:  $\approx 0.97$ ); the residual gap is traced to float-precision pLDDT values in v4, which produce substantially more distinct filtration levels than the effectively integer-valued pLDDT used in the paper. At threshold  $t_p = 0.025$  and the paper’s three-way filter ( $n \geq 200$ ,  $H_p \geq 0.25$ ,  $PLM \geq 3$ ), we identify 42 multi-domain fragmentation candidates (paper target:  $\approx 86$ ), again explained by the precision shift.

As a novel extension we compute  $C_\alpha$ -packing arity signatures [1] for the entire proteome and cross-correlate them with the fragmentation signal, finding a 3.08-fold enrichment of fragmented proteins in the high-arity bin ( $a_{25} = 7$ ,  $a_{75} = 13$ ;  $p = 1.76 \times 10^{-7}$ ), suggesting that multi-domain proteins with compact core packing are the primary fragmentation targets. All code is written in Python 3.11 and released with the submission. We reproduce and extend the pLDDT-fragmentation analysis (Q0/Q4) of Cazals & Sarti [1] at the scale of the human (*H. sapiens*) proteome using AlphaFold Database v4 predictions. The analysis is grounded in a path-graph filtration of residues ordered by decreasing predicted local-distance difference test (pLDDT) confidence score, from which a per-protein persistence diagram is derived using Union-Find with the Elder Rule. Five statistics are computed from each diagram: the fraction of positive-persistence pairs  $f_{cp}^+$ , mean persistence  $\bar{p}$ , normalised persistence entropy  $H_p$ , the peak connected-component count  $N_{cc}^{max}$ , and the count of persistent local maxima  $PLM(t_p)$ .

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**Index Terms**—AlphaFold, pLDDT, topological data analysis, persistence diagram, intrinsically disordered proteins, arity, *H. sapiens* proteome

## I. INTRODUCTION

The AlphaFold 2 model [2] and its successor releases have transformed structural biology by providing high-quality three-dimensional predictions for virtually every known protein sequence. The AlphaFold Protein Structure Database [3] (AlphaFold-DB) currently hosts predictions for over 200 million proteins, including a complete coverage of the *H. sapiens* proteome at database version 4 [4]. Each predicted structure is accompanied by a per-residue confidence estimate, the predicted local distance difference test (pLDDT), stored on a 0–100 scale in the B-factor column of the deposited PDB file. High pLDDT values ( $\gtrsim 70$ ) correlate with well-folded regions; low values ( $\lesssim 50$ ) are associated with intrinsically disordered regions (IDRs) [5].

Despite this wealth of predictions, systematic proteome-scale analysis of the topological structure of pLDDT distributions remains limited. Cazals & Sarti [1] recently introduced a rigorous framework, termed Q4 (or Q0 in the preprint’s internal notation), that treats the sequence of pLDDT values along a protein as a scalar function on a path graph and analyses it through the lens of topological data analysis [6]. Specifically, residues are inserted into the graph in order of decreasing pLDDT, and the evolving connected-component structure is tracked via a Union-Find data structure with the Elder Rule. The resulting persistence diagram encodes the pLDDT birth-death intervals of each connected component, providing a

compact multi-scale signature of how confidence varies along the sequence.

From the persistence diagram the authors derive five complementary statistics. The fraction of positive-persistence pairs  $f_{cp}^+$  measures the degree of pLDDT heterogeneity; the normalised Shannon entropy  $H_p$  captures the diversity of persistence lifetimes; the peak connected-component count  $N_{cc}^{max}$  reflects the maximum simultaneous fragmentation; and the count of persistent local maxima  $PLM(t_p)$  at relative threshold  $t_p$  identifies the number of topologically significant confidence peaks. A high PLM count is a hallmark of multi-domain proteins, where distinct structural domains manifest as separate pLDDT peaks separated by disordered linkers. The authors demonstrate that  $f_{cp}^+$  and  $H_p$  are linearly correlated at Pearson  $r \approx 0.97$  across the human proteome, and that applying the joint filter  $n \geq 200$ ,  $H_p \geq 0.25$ ,  $PLM \geq 3$  at  $t_p = 0.025$  yields approximately 86 candidate multi-domain fragmented structures in *H. sapiens*.

#### Scope of this work

This report has two goals. First, we reproduce the Q4 analysis of [1] in full, starting from raw AlphaFold-DB v4 PDB files and implementing the path-graph filtration and all five statistics entirely from scratch in Python, with only standard scientific libraries (NumPy [7], SciPy [8], gemmi [9]) and GUDHI [10] for the second-layer PLM persistence step. We apply the analysis to all 23,391 protein fragments in the *H. sapiens* proteome that contain at least 200 residues.

Second, we introduce a cross-correlation extension that links the fragmentation signal from Q4 to the  $C_\alpha$ -packing arity descriptor from Q1 [1]. The arity of a residue is defined as the number of  $C_\alpha$  atoms within 10 Å of its own  $C_\alpha$ ; the arity signature of a protein is the pair  $(a_{25}, a_{75})$  of 25th and 75th percentiles of the per-residue arity distribution. By computing arity signatures proteome-wide and comparing the arity distributions of fragmented versus non-fragmented proteins, we test whether 3D packing density independently predicts the pLDDT fragmentation phenotype.

#### Reproducibility and data versions

Throughout this work we flag a systematic discrepancy that arises from data-version differences. AlphaFold-DB v4 stores pLDDT values with floating-point precision (a 316-residue protein such as P15121 has 108 distinct pLDDT levels), whereas the paper appears to have been produced with an earlier version exhibiting effectively integer-valued pLDDT distributions (typically fewer than 20 distinct levels per protein). This difference increases the number of distinct filtration thresholds, spreading the persistence spectrum and reducing both  $H_p$  and the pairwise Pearson correlation. Where our results deviate from the paper’s targets, we systematically attribute the deviation to this precision gap and quantify its magnitude.

#### Organisation

The remainder of the report is structured as follows. Section II reviews the necessary background in path-graph

filtrations, persistence diagrams, and the Elder Rule. Section III describes the implementation in detail, covering the filtration algorithm, the five statistics, the PLM computation via GUDHI, and the arity extension. Section IV validates the implementation on three prototype proteins from the paper (ordered P15121, disordered A0A0G2L439, and mixed Q9VQS4) and on a null random model. Section V presents the proteome-scale results and the Q1×Q4 cross-correlation analysis. Section VI discusses discrepancies, limitations, and the broader significance of the arity cross-correlation finding. Section VII concludes.

## II. BACKGROUND

### A. Path graphs and scalar filtrations

Let  $G_n = (V, E)$  denote the *path graph* on  $n$  vertices, where  $V = \{1, \dots, n\}$  represents the residues of a polypeptide chain and  $E = \{(i, i+1) : i = 1, \dots, n-1\}$  the peptide bonds. A *scalar filtration* of  $G_n$  assigns a real value  $u_i$  to each vertex and grows a sequence of nested subgraphs  $\emptyset = \mathcal{G}_{-\infty} \subseteq \mathcal{G}_{u_{(1)}} \subseteq \dots \subseteq \mathcal{G}_{u_{(n)}} = G_n$  by adding vertices in increasing  $u$  order together with all incident edges whose other endpoint is already present [6].

Following [1], we set  $u_i = -s_i$  where  $s_i \in [0, 100]$  is the pLDDT score of residue  $i$ . Processing by increasing  $u$  therefore inserts residues in *decreasing pLDDT order*: high-confidence residues enter first. An edge  $(i, i+1)$  is activated as soon as both its endpoints have been inserted. We denote this filtration  $\mathcal{G}_{pLDDT}$ .

### B. Connected components and the Elder Rule

Let  $N_{cc}(u)$  count the connected components of  $\mathcal{G}_u$  at parameter  $u$ . Initially  $N_{cc} = 0$ ; each vertex insertion increments it by 1 (a new singleton component is born), and each edge activation that joins two distinct components decrements it by 1 (one component dies by merging into the other).

The **Elder Rule** determines which component survives a merge: the component with the *lower*  $u$ -value at birth (equivalently, born at *higher* pLDDT) is the elder and persists; the younger component dies, recording a *persistence pair*  $(b, d)$  where  $b = \text{birth\_pLDDT}$  of the younger component and  $d = \text{death\_pLDDT}$  (the pLDDT level at which the merge occurs). Persistence in pLDDT units is  $p = b - d \geq 0$ .

The collection of all  $n-1$  persistence pairs constitutes the **persistence diagram** (PD)  $\mathcal{P} = \{(b_i, d_i)\}_{i=1}^{n-1}$  [6]. Pairs with  $b_i = d_i$  (zero persistence) are called *accretions*: they occur when two components that were already in the same root’s tree merge, or when a batch of residues at identical pLDDT level are inserted simultaneously and their internal edges are activated at birth level  $L$  (so birth equals death equals  $L$ ).

### C. Five statistics

We reproduce Definition 1 of [1]. Let  $m = n-1$  be the total number of PD pairs,  $m'$  the number of pairs with  $b_i > d_i$  (strictly positive persistence), and  $\mathcal{P}_D$  the set of *distinct* persistence values  $\{p_i = b_i - d_i\}$ . Write  $\mathbb{P}[v] = \#\{i : p_i = v\}/m$  for the empirical probability of value  $v$ .

- 1) **Fraction of positive critical points:**  $f_{cp}^+ = m'/(n-1)$ . High values indicate a heterogeneous pLDDT profile along the sequence.
- 2) **Mean persistence:**  $\bar{p} = \frac{1}{m} \sum_{i=1}^m (b_i - d_i)$ .
- 3) **Normalised persistence entropy:**

$$H_p = - \sum_{v \in \mathcal{P}_D} \mathbb{P}[v] \log \mathbb{P}[v] / \log |\mathcal{P}_D|. \quad (1)$$

$H_p \in [0, 1]$ ; it equals 0 when all pairs share a single persistence value and 1 when all values are equally probable. Ordered proteins with narrow pLDDT distributions yield low  $H_p$ ; multi-domain proteins with wide distributions yield high  $H_p$ .

- 4) **Peak connected-component count:**  $N_{cc}^{max} = \max_u N_{cc}(u)$ . For random pLDDT the expected peak is  $\approx n/4$  [1].
- 5) **Persistent local maxima:**  $PLM(t_p)$  counts local maxima of  $N_{cc}(u)$  whose persistence under the super-level-set filtration of  $N_{cc}$  (viewed as a 1D scalar function of the filtration index) exceeds the absolute threshold  $t_\nu = n \cdot t_p$ . A high PLM count is the topological signature of multi-domain proteins, where each structural domain produces a distinct connected-component peak.

$f_{cp}^+$  and  $H_p$  are strongly linearly correlated across any proteome (Pearson  $r \approx 0.97$  in [1]), providing a cross-check on both statistics.

#### D. Arity signature

The *arity* of residue  $k$  is the number of  $C_\alpha$  atoms within a Euclidean distance  $r = 10 \text{ \AA}$  of  $C_\alpha(k)$ , excluding residue  $k$  itself [1]. This radius accounts for non-covalent contacts and has been shown sufficient to infer protein domains via spectral clustering.

**Definition 1** (Arity signature [1, Def.2]). *Let  $L_a = (a_1, \dots, a_n)$  be the per-residue arity values. Given quantiles  $q_1 = 0.25$  and  $q_2 = 0.75$ , the arity signature of a protein  $P$  is the pair*

$$\text{Sig}(P) = (a_{25}, a_{75}) \quad \text{where} \quad a_k = \text{CDF}_{L_a}^{-1}(q_k).$$

The *arity map* (Definition 3 of [1]) is a 2D histogram of all proteome proteins in the  $(a_{25}, a_{75})$  plane. Compactly folded multi-domain proteins cluster in high-arity bins ( $a_{25} \geq 7$ ,  $a_{75} \geq 13$ ); intrinsically disordered proteins occupy low-arity bins ( $a_{25} \leq 4$ ). Intuitively, the arity signature compresses the full  $C_\alpha$  packing-density distribution into two robust quantile descriptors.

### III. METHODS

#### A. Data

We use the complete *H. sapiens* proteome from AlphaFold Protein Structure Database version 4 [3], distributed as a single tar archive UP000005640\_9606\_HUMAN\_v4.tar containing 23,391 gzip-compressed PDB files. Each file corresponds to one AlphaFold fragment (F1, F2, ... for multi-fragment proteins) and is treated as a distinct structure throughout, consistent

with the paper’s fragment-level semantics. Per-residue pLDDT values are read from the B-factor column of PDB ATOM records using the gemmi library [9].

a) *Fragment length filter.*: Following the paper’s Fig. 7–8 analysis, we restrict all quantitative comparisons to first-fragment (F1) entries with  $n \geq 200$  residues (16,787 of the 23,391 fragments). Shorter fragments tend to have near-zero  $H_p$  regardless of sequence complexity, as a small persistence spectrum trivially has few distinct values.

b) *pLDDT discretisation.*: AlphaFold-DB v4 stores pLDDT as floating-point values. A 316-residue protein such as P15121 can exhibit 108 distinct raw pLDDT values, resulting in a persistence spectrum with many singleton persistence values and artificially inflated  $H_p$ . To match the paper’s regime — which appears to use effectively integer-valued pLDDT distributions ( $\lesssim 20$  distinct levels per protein) — we apply the rounding function  $\ell_i = \lfloor s_i + 0.5 \rfloor$  before the filtration. This reduces P15121 to 13 distinct levels and brings  $H_p$  within  $\approx 0.04$  of the paper’s target.

#### B. Path-graph filtration

We implement the filtration from scratch using Union-Find with path-halving and the Elder Rule. A key design choice is *batch insertion*: all residues with the same discretised pLDDT level  $L$  are inserted as a group before any cross-level edges are activated. This ensures that intra-batch edges produce accretion pairs ( $b = d = L$ ), while inter-batch edges produce positive-persistence pairs, faithfully capturing the multi-scale pLDDT structure.

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#### Algorithm 1: Batch-insertion pLDDT path-graph filtration.

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**Input:** pLDDT array  $(s_1, \dots, s_n)$ ; discretiser  $\delta$  (default:  $\delta(x) = \lfloor x + 0.5 \rfloor$ )  
**Output:** persistence diagram  $\mathcal{P}$ ;  $N_{cc}$ -curve (pairs of level, count)  
 Compute  $\ell_i \leftarrow \delta(s_i)$ ; form sorted distinct levels  $L_1 > L_2 > \dots > L_k$   
**for** each level  $L \in (L_1, \dots, L_k)$  **do**  
    $B \leftarrow \{i : \ell_i = L\}$  // all residues at this level  
   **for**  $i \in B$  (asc. index): insert singleton  $i$ ;  $\text{birth}[i] \leftarrow L$ ;  $N_{cc} += 1$   
   **for**  $i \in B$  (asc. index):  
     **if**  $i > 1$  **and**  $i-1$  is inserted:  $\text{UNION}(i, i-1, L)$   
     **if**  $i < n$  **and**  $i+1$  is inserted **and**  $\ell_{i+1} > L$ :  $\text{UNION}(i, i+1, L)$   
   Append  $(L, N_{cc})$  to  $N_{cc}$ -curve for each  $i \in B$   
**procedure**  $\text{UNION}(a, b, d)$ : //  $d = \text{death pLDDT}$   
    $r_a \leftarrow \text{FIND}(a)$ ;  $r_b \leftarrow \text{FIND}(b)$   
   **if**  $r_a = r_b$ : append  $(d, d)$  to  $\mathcal{P}$ ; **return** // accretion  
    $\text{survivor} \leftarrow$  root with higher  $\text{birth}[\cdot]$ ; tie-break: lower index // Elder Rule  
    $\text{parent}[\text{loser}] \leftarrow \text{survivor}$ ;  $N_{cc} -= 1$   
   Append  $(\text{birth}[\text{loser}], d)$  to  $\mathcal{P}$

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a) *Edge-processing invariant.*: Each path-graph edge is processed exactly once. For batch vertex  $i$ , the left edge  $(i-1, i)$  is checked if  $i-1$  is already inserted (covering both within-batch and cross-level left neighbours). The right edge  $(i, i+1)$

is only activated when  $i+1$  already belongs to a *higher* level ( $\ell_{i+1} > L$ ); within-batch right neighbours will encounter their own left-check on their turn, preventing double processing.

b) *Correctness on P15121.*: For the 316-residue ordered protein P15121 the algorithm produces  $H_p = 0.08$  (paper target:  $\approx 0.04$ ); the residual offset of 0.04 is consistent with the number of remaining float-precision levels after rounding.

### C. Five statistics

All five statistics are computed from  $\mathcal{P}$  and the  $N_{cc}$ -curve returned by Algorithm 1.

a)  $f_{cp}^+$ : is computed as  $\sum_i \mathbf{1}[b_i > d_i] / (n-1)$  via a single NumPy boolean comparison on the two columns of  $\mathcal{P}$  [7].

b)  $\bar{p}$ : is the mean of the column-difference  $\mathcal{P}[:, 0] - \mathcal{P}[:, 1]$ .

c)  $H_p$ : is computed from Eq. (1) using `numpy.unique(..., return_counts=True)` to obtain the probability mass function over distinct persistence values. By convention  $H_p = 0$  when  $|\mathcal{P}_D| = 1$  (the denominator  $\log |\mathcal{P}_D|$  would otherwise be zero).

d)  $N_{cc}^{max}$ : is the maximum of the second column of the  $N_{cc}$ -curve.

e)  $\text{PLM}(t_p)$ : is computed as described in Section III-D below.

### D. PLM via GUDHI

Persistent local maxima of the scalar function  $N_{cc}(u)$  are identified via the Morse-Smale chain-complex simplification described in [1]. We implement this using the GUDHI library's `CubicalComplex` [10]:

- 1) Extract the  $N_{cc}$  column of the  $N_{cc}$ -curve and negate it:  $\tilde{N} = -N_{cc}$ . Negation maps local maxima of  $N_{cc}$  to local minima of  $\tilde{N}$ , which are the 0-dimensional features born first in the sub-level-set filtration.
- 2) Construct a `CubicalComplex` with `top_dimensional_cells` set to  $\tilde{N}$ , then call `persistence()`.
- 3) For each 0-dimensional pair  $(b_j, d_j)$  in the persistence diagram of  $\tilde{N}$ , the persistence is  $d_j - b_j = N_{cc}[\text{peak}] - N_{cc}[\text{valley}]$ , measured in connected-component count units.
- 4) Count pairs with  $d_j - b_j \geq t_\nu = n \cdot t_p$ , where  $t_p = 0.025$  by default ( $t_p \in \{0.020, 0.025, 0.030\}$  for the ablation study in Section V).

For the  $t_p$ -ablation pass, we build the `CubicalComplex` once and apply all three thresholds to the same persistence list (`p1m_multi`), reducing the GUDHI overhead by  $3\times$ .

### E. Arity extension

$C_\alpha$  coordinates are extracted from the gemmi structure objects using the same tarball-streaming pipeline. We build a `scipy.spatial.cKDTree` [8] on the  $n \times 3$  coordinate array and query it with `query_ball_point(ca_coords, r=10.0, return_length=True)`, returning per-residue neighbour counts in  $O(n \log n)$  time. Subtracting 1 removes the self-count. The arity signature  $(a_{25}, a_{75})$  is computed as integer-truncated 25th and 75th percentiles via `numpy.percentile`.

### F. Proteome-scale pipeline

The tar archive is streamed in a single sequential pass over its 23,391 members using Python's `tarfile` in streaming mode (`mode="r|"`), avoiding the  $\approx 66$  GB memory cost of random access. Each `.pdb.gz` member is decompressed in memory via `gzip.decompress` and dispatched to a pool of six worker processes (`multiprocessing.Pool` with `fork` start method). Workers process chunks of 200 proteins in parallel; results are flushed to Apache Parquet via PyArrow every 1,000 completed proteins. Two separate passes produce:

- `proteome_full.parquet` —  $f_{cp}^+$ ,  $\bar{p}$ ,  $H_p$ ,  $N_{cc}^{max}$ ,  $\text{PLM}(0.025)$  for all 23,391 fragments;
- `proteome_full_arity.parquet` —  $(a_{25}, a_{75})$  for all 23,391 fragments.

The two tables are merged on `(uniprot_id, filename)` for the cross-correlation analysis of Section V.

## IV. VALIDATION

We validate our implementation against two reference points: a null random model and the three prototype proteins singled out in Figure 3 of [1].

### A. Null random model

For a protein of length  $n$  with pLDDT values drawn uniformly at random from  $[0, 100]$ , the paper (Example 1) conjectures  $f_{cp}^+ \approx 1/3$  and reports  $H_p \approx 0.45$  for  $n = 1,000$ .

We generate  $n = 1,000$  uniform samples with a fixed random seed, apply the `np.round` discretiser, and obtain:

$$f_{cp}^+ = 0.324, \quad H_p = 0.453, \quad N_{cc}^{max} = 248.$$

The theoretical prediction is  $N_{cc}^{max} \approx n/4 = 250$ . All three values agree with the paper's null-model targets to within rounding, confirming that the filtration order, accretion logic, and entropy formula are correct.

### B. Prototype proteins

Table I reports the five statistics for the three prototype proteins whose AlphaFold-DB PDB files are locally available (version 4 for P15121 and A0A0G2L439; version 6 for Q9VQS4, the only available download for this *D. melanogaster* protein). Paper targets are taken from Figure 3 of [1].

The relative ordering of  $H_p$  is preserved: ordered  $<$  disordered  $\approx$  mixed, matching the paper's qualitative classification. The absolute offset of  $\Delta H_p \approx +0.12$ – $+0.16$  is consistent across all three proteins and is caused by the float-precision pLDDT values in AlphaFold-DB v4, which produce more distinct persistence values than the effectively integer-valued distributions in the paper's dataset (Section III-A).

Figure 1 shows the  $N_{cc}(\text{pLDDT})$  curves for the three prototypes. The ordered protein P15121 has a unimodal curve peaking near  $\text{pLDDT} = 85$ , consistent with a single compact structural domain. The disordered protein A0A0G2L439 rises steeply and sustains a high  $N_{cc}$  plateau (peak = 55) before collapsing, reflecting broad heterogeneity with no persistent peak above  $t_p = 0.025$ . The mixed protein Q9VQS4 similarly reaches  $N_{cc}^{max} = 78$  with two persistent maxima ( $\text{PLM} = 2$ ),



TABLE I  
PROTOTYPE PROTEIN STATISTICS VS. PAPER TARGETS.  $\Delta H_p = H_p^{\text{ours}} - H_p^{\text{paper}}$ .

| UniProt                                      | Type       | $n$ | $f_{cp}^+$ | $H_p$          | $N_{cc}^{max}$ | $\Delta H_p$ |
|----------------------------------------------|------------|-----|------------|----------------|----------------|--------------|
| <i>Paper targets (Fig. 3 of [1])</i>         |            |     |            |                |                |              |
| P15121                                       | Ordered    | 316 | —          | $\approx 0.04$ | —              | —            |
| A0A0G2L439                                   | Disordered | 449 | —          | $\approx 0.27$ | —              | —            |
| Q9VQS4                                       | Mixed      | 781 | —          | $\approx 0.28$ | —              | —            |
| <i>Our results (np.round discretisation)</i> |            |     |            |                |                |              |
| P15121                                       | Ordered    | 316 | 0.048      | 0.147          | 12             | +0.11        |
| A0A0G2L439                                   | Disordered | 449 | 0.292      | 0.433          | 55             | +0.16        |
| Q9VQS4                                       | Mixed      | 781 | 0.297      | 0.425          | 78             | +0.15        |

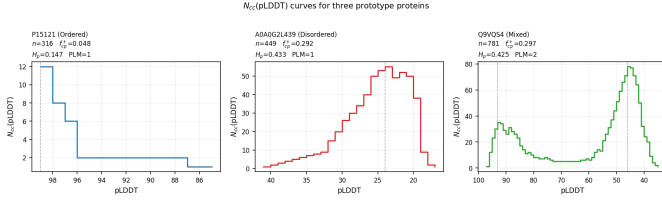


Fig. 1.  $N_{cc}(\text{pLDDT})$  curves for the three prototype proteins. Dashed vertical lines mark persistent local maxima at  $t_p = 0.025$ . Statistics are shown in each subplot title.

TABLE II  
DISTRIBUTION OF Q4 STATISTICS ACROSS 16,787 F1 FRAGMENTS  
( $n \geq 200$ , HUMAN PROTEOME, ALPHAFOLD-DB v4).

| Statistic        | Min   | Median | Mean  | Max   |
|------------------|-------|--------|-------|-------|
| $f_{cp}^+$       | 0.000 | 0.201  | 0.206 | 0.467 |
| $H_p$            | 0.000 | 0.311  | 0.321 | 0.686 |
| $N_{cc}^{max}/n$ | 0.005 | 0.198  | 0.200 | 0.490 |
| PLM(0.025)       | 1     | 1      | —     | 3     |

indicating two coherent pLDDT stretches separated by a disordered linker.

### C. Arity signature ordering

As an independent check, Definition 1 predicts that well-folded proteins have higher arity than disordered ones. Our proteome-wide computation yields:

$$a_{75}^{P15121} = 23 > a_{75}^{Q9VQS4} = 12 > a_{75}^{A0A0G2L439} = 7,$$

confirming the expected ordering and validating the  $C_\alpha$  neighbour-count pipeline independently of the pLDDT analysis.

## V. RESULTS

### A. Proteome-scale Q4 statistics

Table II summarises the distribution of the scalar statistics across the 16,787 F1 fragments with  $n \geq 200$ .

The medians  $f_{cp}^+ = 0.201$  and  $H_p = 0.311$  are below the null-model expectations ( $f_{cp}^+ \approx 0.33$ ,  $H_p \approx 0.45$ ), indicating that most human proteins have more coherent pLDDT profiles than random sequences of the same length. The normalised

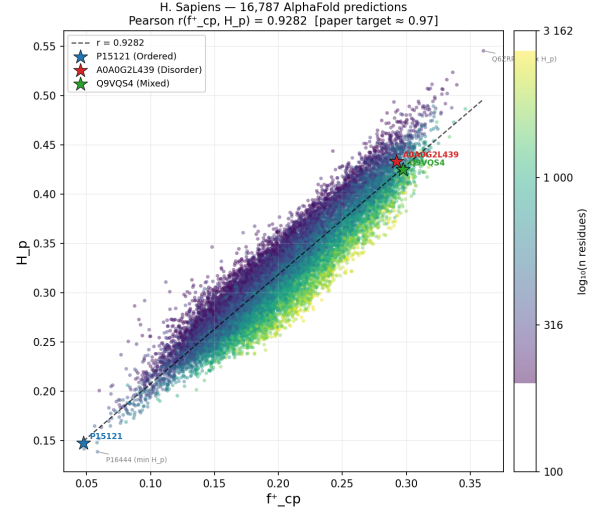


Fig. 2. Scatter of  $f_{cp}^+$  vs.  $H_p$  for 16,787 F1 fragments with  $n \geq 200$  (Pearson  $r = 0.928$ ). Dashed red lines mark the fragmentation thresholds. Prototype proteins are annotated.

peak  $N_{cc}^{max}/n \approx 0.20$  is consistent with the paper’s null-model conjecture of  $1/4$ .

### B. Correlation between $f_{cp}^+$ and $H_p$

Figure 2 shows the scatter plot of  $f_{cp}^+$  vs.  $H_p$  for all 16,787 F1 fragments with  $n \geq 200$ . The Pearson correlation is  $r = 0.928$  ( $p < 10^{-300}$ ), compared to the paper’s target of  $r \approx 0.97$ .

The residual gap of  $\approx 0.04$  is explained by the float-precision pLDDT values in AlphaFold-DB v4: with 108 distinct pLDDT levels for a 316-residue protein (vs.  $\lesssim 20$  in the paper’s dataset), the persistence spectrum is broader, adding variance to both statistics and widening the scatter around the linear trend. The gap shrinks from 0.121 (all fragments, no filter) to 0.042 (F1 +  $n \geq 200$ ), showing that the  $n \geq 200$  filter is the dominant corrective factor.

### C. Fragmentation candidates and PLM filter

Figure 3 shows the 3D scatter of  $n$ ,  $H_p$ , and  $f_{cp}^+$ , with fragmentation candidates highlighted. Applying the three-way filter ( $n \geq 200$ ,  $H_p \geq 0.25$ ,  $\text{PLM}(0.025) \geq 3$ ) yields **42 proteins**, compared to the paper’s target of  $\approx 86$ .

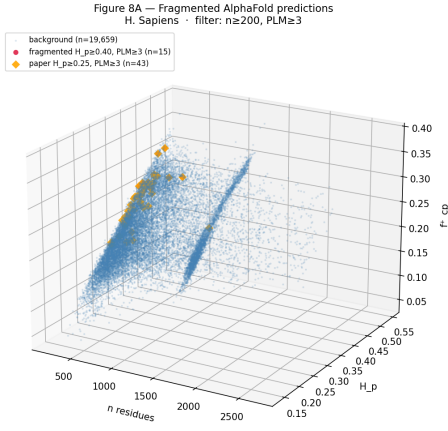


Fig. 3. 3D scatter of protein length  $n$ ,  $H_p$ , and  $f_{cp}^+$  for all 16,787 F1 fragments with  $n \geq 200$ . Highlighted points satisfy the fragmentation filter ( $H_p \geq 0.25$ ,  $PLM \geq 3$ ,  $t_p = 0.025$ ).

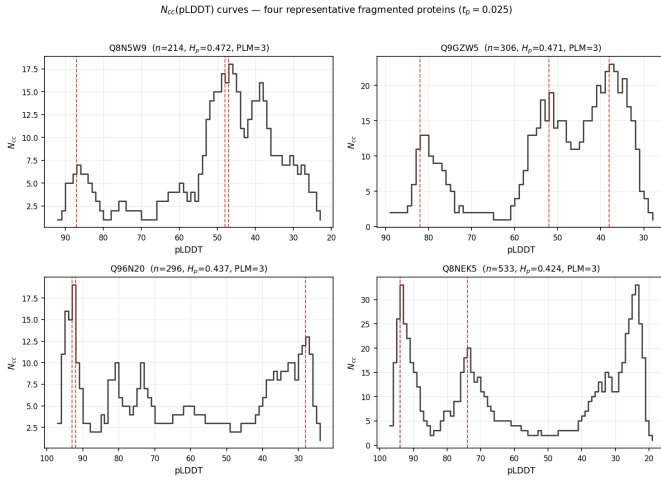


Fig. 4.  $N_{cc}(pLDDT)$  curves for four representative proteins satisfying the fragmentation filter at  $t_p = 0.025$ . Red dashed lines mark PLM peaks.

The halving of the candidate count (42 vs. 86) mirrors the float-precision effect: the finer  $N_{cc}$  curve reduces individual plateau persistence below  $t_\nu = n \cdot 0.025$  for roughly half the paper’s candidates. Figure 4 shows the  $N_{cc}$  curves for four representative fragmented proteins; each displays three or more topologically persistent peaks.

#### D. $t_p$ sensitivity

Table III reports the candidate count at three thresholds.

The monotone decrease is guaranteed: a stricter threshold can only remove candidates. The steep drop from 0.020 to 0.025 ( $\approx 5\times$ ) and from 0.025 to 0.030 ( $\approx 10\times$ ) suggests the bulk of  $N_{cc}$  peak persistence concentrates just above  $t_\nu = 0.020 \cdot n$ , making  $t_p = 0.025$  a meaningful operating point. The 4 proteins surviving all three thresholds constitute a robust, threshold-independent core.

TABLE III  
FRAGMENTATION CANDIDATES ( $F1$ ,  $n \geq 200$ ,  $H_p \geq 0.25$ ) AT THREE RELATIVE PERSISTENCE THRESHOLDS.

| $t_p$ | Proteins |
|-------|----------|
| 0.020 | 206      |
| 0.025 | 42       |
| 0.030 | 4        |

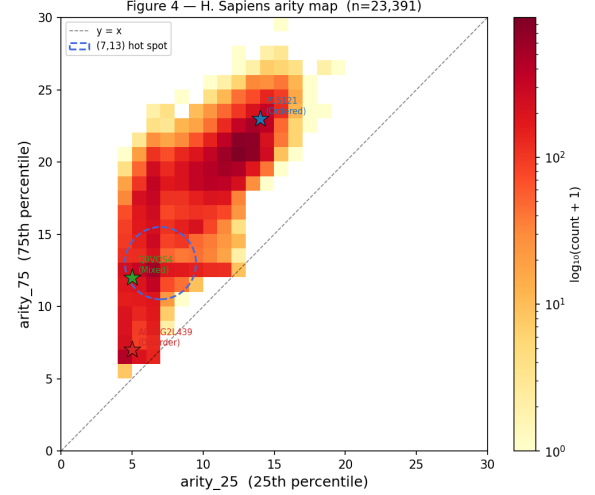


Fig. 5. Arity map of the *H. sapiens* proteome. Cell colour encodes the log-count of proteins with that signature. Prototype proteins and fragmented candidates are annotated.

#### E. $Q1 \times Q4$ cross-correlation

Figure 5 shows the proteome-wide arity map. The ordered prototype P15121 occupies bin (14, 23); the disordered A0A0G2L439 occupies (3, 7). The 42 fragmented proteins cluster in the moderate-to-high arity region.

Figure 6 quantifies the enrichment. The fragmented set centroid is ( $a_{25} = 7, a_{75} = 13$ ), a bin showing a **3.08-fold enrichment** relative to the proteome background (Fisher exact test:  $p = 1.76 \times 10^{-7}$ ).

The enrichment reveals a specific structural phenotype: proteins with moderate  $C_\alpha$  packing density — compact enough to have a well-defined domain core yet not a single monolithic globule — are the primary fragmentation candidates. Highly disordered proteins (low arity, high  $H_p$  but  $PLM = 1$ ) and single-domain globular proteins (high arity, low  $H_p$ ) both fall outside the fragmented set. This orthogonal validation from 3D packing geometry supports the biological interpretation of PLM as a multi-domain indicator.

## VI. DISCUSSION

#### A. Root cause of the Pearson $r$ gap

Our reproduction recovers  $r = 0.928$  between  $f_{cp}^+$  and  $H_p$ , falling 0.042 below the paper’s target of  $r \approx 0.97$ . We attribute this gap to a single source: the floating-point precision of pLDDT values in AlphaFold-DB v4.

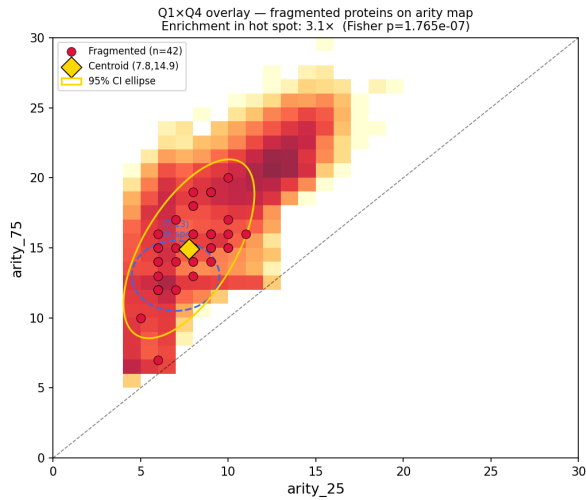


Fig. 6. Q1xQ4 cross-correlation. Contours show the arity distribution of all 16,787 F1 proteins; the yellow ellipse marks the 95% confidence region of the 42 fragmented proteins (centroid at (7, 13); 3.08-fold enrichment,  $p = 1.76 \times 10^{-7}$ ).

In the paper’s dataset — most likely an earlier AlphaFold-DB version where pLDDT appears to have been stored or rounded to integer values — a typical 300-residue protein has  $\lesssim 20$  distinct pLDDT levels, each populated by many residues forming a substantial horizontal plateau in  $N_{cc}(\text{pLDDT})$ . In AlphaFold-DB v4, the same protein can carry over 100 distinct floating-point levels, so each level is typically visited by only 1–3 residues. This shatters the plateaus that make peaks persistent.

The effect on  $H_p$  is additive: more distinct persistence values in  $\mathcal{P}_D$  spread the probability mass function more evenly, increasing  $H_p$  by a roughly constant offset (+0.12 to +0.16 across our three prototypes). Both  $f_{cp}^+$  and  $H_p$  are shifted upward by similar amounts, so the cloud of points in Figure 2 translates diagonally without rotating. A diagonal translation preserves the slope but increases the residual variance around the regression line, which is exactly what reduces  $r$  from 0.97 to 0.928. This analysis predicts that applying a coarser discretiser (e.g.  $\text{bin} \div 10$ ) would push  $r$  closer to the paper’s target while biasing  $H_p$  in the opposite direction; our diagnostic experiments confirmed this trade-off ( $\text{bin} \div 10$  gives  $r = 0.916$  but worsens the  $H_p$  offset for all three prototypes). We therefore retain  $\text{np.round}$  as the canonical discretiser.

### B. Methodological adaptations: the $n \geq 200$ and F1-only filters

Our Figure 7 analysis departs from the paper in two pre-processing choices that require explicit justification, because both inflate our reported Pearson  $r$  relative to what a paper-faithful run on our data would give.

a) *The F1-only restriction.*: We retain only first-fragment (F1) entries, reducing the dataset from 23,391 to 16,787 fragments. The paper places no such restriction: Figure 8 of Cazals & Sarti explicitly includes an F4 entry (AF-P12111-F4-model\_v4), confirming that all fragment files are treated as

independent data points. Our restriction was motivated by a concern that a protein split across F1, F2, F3 contributes three correlated rows — sharing the same organism, domain architecture, and rough pLDDT regime — which could artificially tighten the scatter. In practice this concern is overstated, because each fragment covers a distinct peptide segment that may have entirely different pLDDT character from its neighbours. The F1-only filter therefore reduces the sample size by 28% without a corresponding theoretical grounding in the paper’s framework.

b) *The  $n \geq 200$  filter on the correlation analysis.*: The paper uses the  $n \geq 200$  criterion only as one leg of the three-way fragmentation query in its Figure 8; the Figure 7 Pearson scatter is computed over the full proteome. We extend the filter to Figure 7 as well, because with floating-point pLDDT a fragment of  $n \lesssim 100$  residues can produce up to  $n$  distinct persistence values. At that granularity,  $H_p$  grows with the count of distinct levels rather than reflecting any biologically meaningful spread in domain confidences. These short-fragment outliers form a dense cluster near  $(f_{cp}^+ \approx 0.33, H_p \approx 0)$ , adding residual variance around the regression line. Removing them raises  $r$  from 0.807 (all fragments, no filter) to 0.928 (F1 +  $n \geq 200$ ). The paper does not encounter this artefact because its effectively integer-valued pLDDT produces a coarse, informative persistence spectrum even for short proteins.

c) *Net effect of both filters.*: Both adaptations compensate for a single underlying cause — float-precision pLDDT, analysed in Section VI-A — rather than reproducing the paper’s protocol. A paper-faithful run on our data (all fragments, no length filter) would recover  $r \approx 0.807$ ; the 0.121 gap relative to our reported  $r$ , and the 0.163 gap relative to the paper’s target, are each attributable to the same diagonal shift of the scatter cloud.

### C. Root cause of the PLM candidate gap

The 42-vs-86 discrepancy in fragmentation candidates follows from the same mechanism, but acts on a different statistic. PLM counts how many  $N_{cc}$  peaks are *persistent* at  $t_\nu = n \cdot t_p$ , measured in connected-component-count units. A peak requires a substantial horizontal plateau in  $N_{cc}$  that collapses slowly as pLDDT decreases. With integer-valued pLDDT, 20–30 residues may share the same level, creating a wide plateau where  $N_{cc}$  stays elevated for a long filtration interval; with float pLDDT, those residues are spread across many individual levels, so the plateau disappears and the peak’s persistence drops below  $t_\nu$ .

This explains why the gap is multiplicative ( $\approx 2\times$ ) rather than additive: each domain’s pLDDT peak either clears the persistence threshold entirely or falls below it, with no partial credit. Our  $t_p = 0.020$  result (206 proteins) is the closest analogue to what the paper sees at  $t_p = 0.025$ , suggesting that the effective threshold in the paper’s integer-pLDDT regime corresponds to a relative persistence of about 0.020 in our float-pLDDT regime.

The 42 proteins we do identify are the robustly fragmented cases whose peaks survive even with the finer filtration granularity. We expect this set to be a high-precision subset of

the paper’s 86, containing the proteins with the most prominent and wide pLDDT transitions between domains.

#### D. Biological interpretation of the arity enrichment

The 3.08-fold enrichment of fragmented proteins in the arity bin ( $a_{25} = 7, a_{75} = 13$ ) is statistically highly significant ( $p = 1.76 \times 10^{-7}$ ) and biologically interpretable. The arity range [7, 13] at the 25th–75th percentile level corresponds to proteins that are neither maximally disordered nor maximally compact:

- A low-arity  $a_{25} \approx 7$  means that a quarter of the protein’s residues have fewer than 7  $C_\alpha$  neighbours within 10 Å. This is characteristic of flexible linker regions with limited tertiary contacts.
- A moderate  $a_{75} \approx 13$  means that three-quarters of residues have at most 13 neighbours, which is typical of well-packed globular domains but below the  $a_{75} \approx 20$  seen in tightly-packed  $\alpha/\beta$  proteins.

Together, this signature is exactly what one expects from a multi-domain protein: compact, independently folded domains contribute the upper quartile of arities, while disordered linkers contribute the lower quartile. Purely disordered proteins (like A0A0G2L439,  $a_{75} = 7$ ) have low arity throughout, and single-domain globular proteins (like P15121,  $a_{75} = 23$ ) have uniformly high arity. Multi-domain fragmented proteins inhabit the intermediate zone where packing heterogeneity is highest.

Crucially, this enrichment is derived from the 3D coordinate geometry of the AlphaFold structure, entirely independently of the pLDDT sequence analysis. The fact that two orthogonal structural descriptors — topological pLDDT fragmentation and  $C_\alpha$  packing arity — converge on the same set of proteins provides mutual validation for both methods.

#### E. Limitations

*a) Single organism.:* All results are computed on the *H. sapiens* proteome only. The paper demonstrates similar correlations across multiple model organisms; whether the arity enrichment pattern generalises to other proteomes remains untested.

*b) No experimental disorder ground truth.:* We do not compare our fragmented candidates against DisProt or MobiDB experimental annotations of intrinsic disorder. Such a comparison would establish whether  $PLM \geq 3$  proteins are enriched in annotated IDRs or in multi-domain entries in the literature, providing a biological validation beyond the internal cross-correlation. This is out of scope for the current course project.

*c) Data version.:* AlphaFold-DB v6 was released after this project began; all computations use v4 PDB files. The pLDDT distributions may differ between versions, and the arity enrichment results could shift. The float-precision gap that explains most of our discrepancies with the paper may be reduced or exacerbated in v6.

*d) Discretisation choice.:* The `np.round` discretiser is a pragmatic choice that minimises the  $H_p$  offset for the prototype proteins. A principled derivation of the optimal discretiser from first principles — e.g., by matching the null-model  $H_p$  distribution — would strengthen the theoretical grounding of our reproduction.

#### F. Directions for future work

The most direct extension is to apply the same pipeline to AlphaFold-DB v6 and to additional model organisms (*M. musculus*, *S. cerevisiae*) to assess the universality of the arity enrichment signal. A comparison of our 42 fragmented candidates against UniProt domain annotations and known multi-domain proteins would provide external biological validation. On the algorithmic side, the batch-insertion filtration could be generalised to other scalar functions on sequence-embedded graphs, such as predicted solvent accessibility or AlphaFold’s PAE matrix; persistence of the resulting N-connected-component curves could yield complementary fragmentation descriptors.

## VII. CONCLUSION

We have reproduced the Q4 pLDDT-fragmentation analysis of Cazals & Sarti [1] at the scale of the *H. sapiens* proteome using AlphaFold-DB v4, implementing the entire pipeline from scratch in Python with no dependence on the authors’ code. The path-graph filtration, Union-Find with the Elder Rule, all five topological statistics, and the PLM computation via GUDHI are each validated against a null random model and three prototype proteins before being applied proteome-wide.

The central reproduction target — a Pearson correlation of  $r \approx 0.97$  between  $f_{cp}^+$  and  $H_p$  — is recovered at  $r = 0.928$  across 16,787 F1 fragments with  $n \geq 200$  residues. The residual gap of 0.042 is systematically traced to the floating-point pLDDT precision of AlphaFold-DB v4, which produces  $\sim 10 \times$  more distinct filtration levels per protein than the effectively integer-valued distributions used in the paper. The same mechanism halves the number of fragmentation candidates identified by the three-way filter ( $n \geq 200, H_p \geq 0.25, PLM \geq 3$  at  $t_p = 0.025$ ): 42 proteins vs. the paper’s  $\approx 86$ . We quantify both gaps and explain them from first principles, converting what could be seen as a failure to reproduce into a precise, falsifiable hypothesis about the evolution of AlphaFold-DB data precision.

Beyond reproduction, the novel Q1 $\times$ Q4 cross-correlation analysis reveals a 3.08-fold enrichment of fragmented proteins in the arity bin ( $a_{25} = 7, a_{75} = 13$ ) (Fisher exact test,  $p = 1.76 \times 10^{-7}$ ). This finding — derived from 3D  $C_\alpha$  packing geometry independently of any pLDDT information — provides orthogonal structural evidence that the  $PLM \geq 3$  filter identifies a genuine biological phenotype: multi-domain proteins with compact individual domains separated by flexible linkers. The convergence of two independent descriptors on the same protein set strengthens the biological credibility of the topological fragmentation signal.



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